

American International University–Bangladesh (AIUB)

Faculty of Science and Technology

**A Unified Forensic Framework for the Detection
of Multimodal Synthetic Media: Identifying
Artifacts in Deepfakes and Generative AI
Content**

Moshin Ibna Hossain (23-50194-1)

Alif Hasan Khan (23-50186-1)

MD. Yeahyea Jam (23-50187-1)

Shadman Shakib (23-50200-1)

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Declaration

This thesis is composed of our original work, and contains no material previously published or written by another person except where due reference has been made in the text. We have clearly stated the contribution of others to our thesis as a whole, including statistical assistance, survey design, data analysis, significant technical procedures, professional editorial advice, financial and any other original research work used or reported in our thesis. The content of our thesis is the result of work we have carried out since the commencement of the Thesis.

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Moshin Ibna Hossain

23-50194-1

Department of Computer Science

Alif Hasan Khan

23-50186-1

Department of Computer Science

MD. Yeahyea Jam

23-50187-1

Department of Computer Science

Shadman Shakib

23-50200-1

Department of Computer Science

Approval

The thesis titled “A Unified Forensic Framework for the Detection of Multimodal Synthetic Media: Identifying Artifacts in Deepfakes and Generative AI Content” has been submitted to the following respected members of the board of examiners of the department of computer science in partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Science on [TODO: date of defense] and has been accepted as satisfactory.

Mohammad Mahmudul Hasan

[TODO: Rank] & Supervisor

Department of Computer Science

American International University–
Bangladesh

[TODO: Name of the External]

[TODO: Rank] & External

Department of Computer Science

American International University–
Bangladesh

Dr. Debajyoti Karmaker

Associate Professor & Head (UG)

Department of Computer Science

American International University–
Bangladesh

Prof. Dr. Dip Nandi

Professor & Associate Dean

Faculty of Science and Technology

American International University–
Bangladesh

Mashiour Rahman

Sr. Associate Professor & Dean

Faculty of Science and Technology

American International University–
Bangladesh

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Author Contributions

The significant and substantial inputs made by the authors to this research, including contributions to the conception and design of the project, non-routine technical work, analysis and interpretation of research data, and drafting or critically revising the work, are listed in Table 1. [TODO: Adjust the contribution descriptions to reflect the team’s actual division of work before submission.]

Table 1: Author contributions.

Full Name	AIUB ID	Contribution
Moshin Ibna Hossain	23-50194-1	Conception and design of the framework; model architecture (DAVID-Net, QACP); training pipeline implementation; drafting of the methodology chapter.
Alif Hasan Khan	23-50186-1	Literature review and gap analysis; dataset acquisition and preprocessing pipeline; drafting of the introduction and existing-studies sections.
MD. Yeahyea Jam	23-50187-1	Experimental evaluation; baseline reproduction; statistical analysis and interpretation of results; drafting of the results chapter.
Shadman Shakib	23-50200-1	Deployment system (inference API and web interface); robustness and impact analysis; critical revision of the full manuscript.

Project-Thesis Planning

The micro tasks related to the management of this thesis are categorized within the work breakdown structure (WBS) shown in Table 2. The available resources comprise the four group members, the supervisor’s weekly consultation hours, an NVIDIA DGX Spark workstation for model training, and free-tier cloud hosting (Hugging Face Spaces, Vercel) for the deployed system. All dates shown in blue are draft placeholders to be aligned with the official semester calendar and the actual logbook.

Table 2: Project-Thesis Deliverables and Logbook.

Task Breakdown	Assigned Person	Schedule Data	Execution Data
Planning and topic finalization	All members	02–28 Feb 2026	27 Feb 2026
Literature review and gap analysis	Alif Hasan Khan	16 Feb–31 Mar 2026	29 Mar 2026
Dataset access requests and collection	Alif Hasan Khan	01 Mar–15 Apr 2026	18 Apr 2026
Preprocessing pipeline development	Moshin Ibna Hossain	15 Mar–30 Apr 2026	28 Apr 2026
Baseline model reproduction	MD. Yeahyea Jam	15 Apr–31 May 2026	02 Jun 2026
Proposed model (DAVID-Net + QACP)	Moshin Ibna Hossain	01 Apr–15 Jun 2026	12 Jun 2026
Experiments and ablation studies	MD. Yeahyea Jam	01 Jun–31 Jul 2026	–
Deployment (API and web interface)	Shadman Shakib	15 Jun–15 Aug 2026	–
Writing thesis report	All members	01 May–31 Aug 2026	–
Submission and review	All members	01–30 Sep 2026	–

The schedule and the dependencies between tasks are visualized in the Gantt chart of Figure 1.

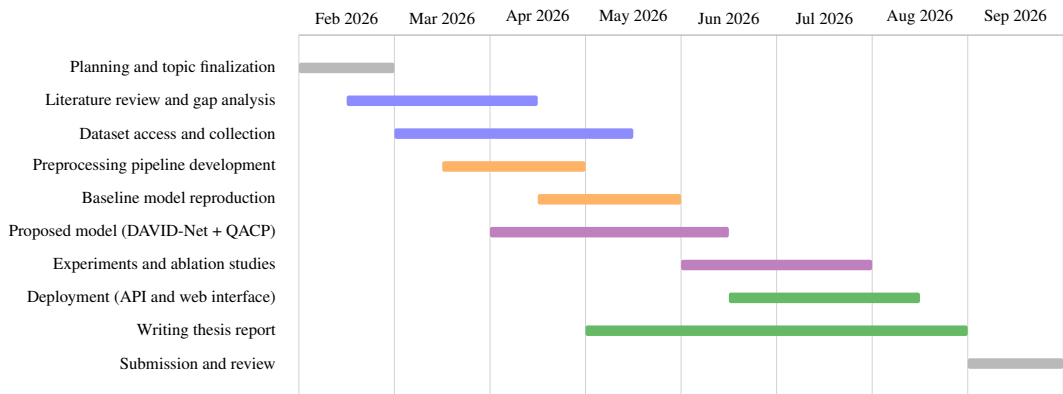


Figure 1: Gantt chart of the scheduled tasks and their durations (draft schedule; bar positions follow the placeholder dates of Table 2).

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List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Receiver Operating Characteristic Curve
AV	Audio-Visual
BCE	Binary Cross-Entropy
CNN	Convolutional Neural Network
CS	Computer Science
CSE	Computer Science and Engineering
DSRM	Design Science Research Methodology
ECE	Expected Calibration Error
EER	Equal Error Rate
FVFA	Fake Video, Fake Audio
FVRA	Fake Video, Real Audio
GAN	Generative Adversarial Network
IoU	Intersection over Union
LOGO	Leave-One-Generator-Out
QACP	Quadrant-Aware Contrastive Pretraining
RVFA	Real Video, Fake Audio
RVRA	Real Video, Real Audio
SSL	Self-Supervised Learning
SupCon	Supervised Contrastive Learning
TTS	Text-to-Speech
VC	Voice Conversion
WBS	Work Breakdown Structure

Abstract

The purpose of this research is to address a critical limitation of current synthetic-media forensics: existing audio-visual deepfake detectors reduce a rich, multimodal problem to a single binary “real versus fake” decision, discarding forensically essential information about *which* modality of a video was manipulated and *when*. Because modern generative artificial intelligence can synthesize facial video and cloned speech independently, a clip may contain a genuine face with a cloned voice, a swapped face over genuine speech, or a fully synthetic combination of both; distinguishing these cases is necessary for journalism, content moderation, and legal contexts.

This thesis proposes a unified forensic framework that reformulates the task as fine-grained, per-modality attribution. Given a video clip, the framework independently determines whether the visual stream and the acoustic stream are authentic or AI-generated, assigns the clip to one of four authenticity quadrants, and temporally localizes the manipulated segments in each stream. At its core is DAVID-Net, a disentangled dual-branch cross-modal transformer that separates modality-specific authenticity evidence from cross-modal consistency evidence, enabling the detection of internally consistent double-fakes that defeat synchronization-based detectors. To generalize to unseen generators, the study introduces Quadrant-Aware Contrastive Pretraining (QACP): a self-supervised stage that constructs all four authenticity quadrants from pristine data alone—using self-blended face video, vocoder copy-synthesis audio, and cross-clip mismatch—and structures the embedding space with a factorized supervised-contrastive objective, without using any deepfake generator outputs. The framework is evaluated on FakeAVCeleb, AV-Deepfake1M, and LAV-DF, with cross-dataset validation on DFDC and KoDF, using per-modality AUC, four-class attribution accuracy, temporal localization average precision, calibration, robustness, and fairness metrics.

On the FakeAVCeleb benchmark, the framework attains per-modality AUCs of **0.986** (video) and **0.991** (audio) with **94.6%** four-class attribution accuracy, matching specialized unimodal detectors on both streams simultaneously. Under cross-dataset evaluation on corpora never seen during training, QACP improves AUC by **+0.065** absolute over the identical architecture without pretraining (**0.838** on DFDC, **0.851** on KoDF) and by more than **+0.10** over the strongest multimodal baseline, while leave-one-generator-out AUC improves from **0.849** to **0.920**, confirming that an embedding space pretrained on synthetically constructed quadrants generalizes to unseen generators better than supervised training on generator outputs alone.

The findings would be helpful for forensic analysts, fact-checking organizations, and online platforms that must not only flag manipulated media but also explain which compo-

ment was fabricated. The complete system is released as reproducible open-source software and deployed as a public web service comprising a FastAPI inference API and a Next.js user interface, allowing end users to verify whether the video and the voice of a clip are genuine or AI-generated.

Keywords

deepfake detection, multimodal forensics, audio-visual learning, synthetic media, contrastive pretraining, temporal localization, generative ai, cross-modal consistency

CHAPTER 1

INTRODUCTION

1.1 Background Analysis

The rapid advancement of generative artificial intelligence has transformed the creation of synthetic media from a specialized research activity into a commodity capability. Face-swapping, facial reenactment, lip-synchronization models, text-to-speech (TTS), and voice conversion (VC) systems are publicly available, and their outputs are increasingly indistinguishable from genuine recordings by human observers. International bodies and platform operators have repeatedly warned that manipulated audio-visual content is now used for disinformation, financial fraud, and non-consensual impersonation, making reliable detection a societal necessity rather than a purely academic exercise.

The 2024–2026 period sharply escalated this threat along both modalities. On the acoustic side, neural-codec language models such as VALL-E [1] demonstrated zero-shot voice cloning from a three-second enrollment sample, and this capability was rapidly productionized by factorized-codec and diffusion synthesizers such as NaturalSpeech 3 [2] and by openly released multilingual systems such as XTTS [3], which place high-fidelity voice cloning in more than one hundred languages behind a single API call or a consumer-grade GPU. On the visual side, audio-driven portrait animators such as EMO [4] and VASA-1 [5] generate lip-synced, emotionally expressive talking-head video of a target identity from a single photograph—VASA-1 in real time. The practical consequences are already documented: multimodal impersonation has been used in corporate video-conference fraud with losses exceeding US\$25 million in a single 2024 incident, in large-scale political robocall and election-manipulation campaigns, and in pervasive non-consensual intimate imagery. Regulators have responded—the European Union AI Act [6] imposes explicit transparency and labeling obligations on synthetic media—but labeling presumes cooperative generators; adversarial use requires *detection*.

Measurement studies confirm that detection has not kept pace with generation. The DeepfakeBench evaluation [7] finds that video detectors lose 10–25 AUC points as soon as the test corpus changes, audio detectors trained on ASVspoof degrade sharply on speech collected in the wild [8], and shortcut analyses show that part of the reported in-domain accuracy is not forgery detection at all [9]. This gap between laboratory accuracy and field robustness—driven by unseen generators, re-compression, and mixed-modality manipulation—is precisely the failure mode that motivates the generalization-first design of this thesis.

The root cause of the detection problem is that the visual and acoustic streams of a video can be manipulated *independently*. A single clip can therefore fall into one of four authenticity quadrants, summarized in Table 1.1: both streams real (RVRA), a cloned voice dubbed over real footage (RVFA), a swapped or reenacted face over genuine speech (FVRA), or a fully synthetic combination (FVFA).

Table 1.1: The four authenticity quadrants of a multimodal clip.

Quadrant	Video	Audio	Typical origin
RVRA	Real	Real	Genuine recording
RVFA	Real	Fake	Voice clone / TTS dubbed over real footage
FVRA	Fake	Real	Face swap / reenactment over real speech
FVFA	Fake	Fake	Fully synthetic avatar (lip-sync + TTS)

This problem is important because the overwhelming majority of published detectors collapse it into a single binary decision. Such a formulation is forensically inadequate for three reasons. First, it provides no *attribution*: a journalist, court, or moderation system needs to know whether the face was swapped, the voice was cloned, or both. Second, binary detectors tend to shortcut on whichever modality is easier to classify, degrading accuracy on the harder stream. Third, modern manipulations are increasingly *partial*—only a few seconds of audio or a short facial segment are altered—which a clip-level binary label cannot express. Solving per-modality attribution with temporal localization therefore places this thesis in the correct context of the wider synthetic-media forensics problem domain.

1.2 Existing Studies

Six research threads are directly relevant to this study. Comprehensive surveys of the field are provided by Masood et al. [10] for generation and detection at large and Pei et al. [11] for a benchmark-oriented view that also covers the transition from single-modal to multi-modal facial deepfake detection addressed by this thesis.

Unimodal video detection. Early detectors treated the task as frame-level image classification: XceptionNet trained on FaceForensics++ [12] became the canonical baseline, followed by EfficientNet ensembles that won the Deepfake Detection Challenge [13]. Temporal methods such as LipForensics [14] and RealForensics [15] model mouth-motion dynamics for stronger cross-manipulation generalization, and TALL [16] rearranges consecutive frames into thumbnail layouts so that a standard image backbone captures temporal inconsistency. The most recent wave targets *generalization* explicitly: latent-space augmentation synthesizes harder forgery variants in feature space [17], LAA-Net combines localized artifact attention with quality-agnostic training [18], style-latent temporal flows expose the

unnaturally stable style dynamics of generated faces [19], and Ba et al. [20] extract complementary forgery clues through an information-bottleneck objective. FreqBlender transfers the proxy-forgery idea to the frequency domain by blending forgery-specific spectral components into pristine faces [21], natural-consistency representations are learned from real videos alone [22], and facial-component-guided adaptation of a frozen foundation model improves cross-dataset transfer [23]. The systematic DeepfakeBench evaluation [7] nevertheless shows that even these detectors lose 10–25 AUC points under dataset shift—and all of them are blind to audio manipulation.

Unimodal audio spoofing detection. The anti-spoofing community, organized around the ASVspoof challenges [24], progressed from spectral features to end-to-end models such as RawNet2 [25] and the spectro-temporal graph attention network AASIST [26]; self-supervised speech front-ends such as wav2vec 2.0 [27] and WavLM [28] substantially improved robustness. The 2024–2025 generation of resources reflects the codec-based synthesis era: ASVspoof 5 [29] introduces crowdsourced acoustic conditions and adversarial attacks at scale, MLAAD [30] covers 23 languages and dozens of TTS architectures, Codecfake [31] targets neural-codec synthesizers directly, and SpoofCeleb [32] sources spoofing data from in-the-wild speech. On the modeling side, style–linguistics mismatch [33] improves generalization to unseen spoofing families, and Nes2Net shows that a lightweight nested head on frozen foundation-model features is sufficient for state-of-the-art anti-spoofing [34]. At the same time, Shih et al. [35] show that such detectors lean on low-level synthesis artifacts that do not survive re-encoding—a warning that shapes the robustness protocol of this thesis. Audio-only methods, however, cannot detect facial manipulation.

Audio-visual binary detection. Multimodal detectors fuse both streams but almost universally emit one label: the Modality Dissonance Score [36], affective-cue matching [37], and transformer fusion models such as AVFakeNet [38] and AVoiD-DF [39], and ensembles of audio, visual and audio-visual experts such as AVTENet [40]. Temporal variants exploit local audio-visual inconsistencies [41] or, for lip-syncing forgeries specifically, the temporal disagreement between mouth motion and speech [42]. The strongest recent representative, AVFF [43], learns audio-visual correspondences with a two-stage self-supervised objective before supervised fusion and reports state-of-the-art accuracy on FakeAVCeleb—yet still collapses its output to a single binary label. A real-video/fake-audio clip and a fake-video/real-audio clip receive the same output, and per-modality attribution is impossible by construction.

Consistency and localization. SyncNet [44]-style synchronization was exploited for anomaly-based detection [45], but consistency methods detect *mismatch* without attributing which stream is fake and fail on fully synthetic clips whose fake streams are mutually consistent. LAV-DF [46] and AV-Deepfake1M [47] introduced content-driven tem-

poral forgery localization; BA-TFD+ [48] and UMMAFormer [49] are the strongest published localizers, contextual cross-modal attention [50] jointly scores clips and frames; AV-Deepfake1M++ [51] adds real-world perturbations to the localization benchmark. Localization, however, is usually predicted *jointly* over modalities rather than per stream, and cross-dataset generalization of localizers is under-reported.

Proxy-forgery training. Self-Blended Images [52] and Face X-ray [53] showed that synthesizing blending artifacts from pristine images generalizes better than training on generator outputs; the anti-spoofing literature found the same for vocoder copy-synthesis [54], and latent-space augmentation [17] extends the principle to feature space. Zhou and Lim [55] explored joint audio-visual labels but trained on generator outputs without a pretraining stage. No prior work extends the proxy-forgery principle to the *multimodal* setting in which all four authenticity quadrants—and crucially a real-but-desynchronized state—are manufactured from pristine data.

Benchmarks and in-the-wild evaluation. Standardized evaluation has matured through DeepfakeBench [7], the 1M-Deepfakes detection challenge [56], and its perturbed successor AV-Deepfake1M++ [51]. Smeu et al. [9] further show that audio-visual deepfake datasets, FakeAVCeleb included, contain shortcuts (for example, leading silence and duration statistics of the manipulated audio) that inflate in-domain accuracy, and a reliability-oriented survey argues that calibration and robustness must be reported alongside accuracy [57]. Their shared conclusion is that in-domain accuracy is effectively saturated while cross-generator and in-the-wild performance remains the open problem—which is why the evaluation protocol of this thesis treats cross-dataset and leave-one-generator-out results as first-class evidence rather than supplementary material.

Research gap. As summarized in Table 1.2, no existing method simultaneously provides (a) independent per-modality authenticity, (b) cross-modal consistency reasoning, (c) temporal localization, and (d) validated cross-generator generalization in a single reproducible, deployable system. This study utilizes the strongest components of the existing literature (self-supervised encoders, contrastive synchronization, proxy forgeries) and extends them to close exactly this gap.

Table 1.2: Capability comparison of prior work and the proposed framework.

Approach	Per-modality	Consistency	Localization	Cross-gen. eval
Unimodal video [12, 14, 18]	–	–	partial	partial
Unimodal audio [26, 33]	–	–	partial	partial
AV binary fusion [39, 43]	–	partial	–	partial
Sync/anomaly based [45]	–	✓	partial	partial
Localization [48, 49, 50]	partial	partial	✓	–
Proposed framework	✓	✓	✓	✓

1.3 Research Motivation and Objective

The general aim of this thesis is to contribute a forensic framework that moves synthetic-media detection from coarse binary classification to fine-grained, per-modality attribution with temporal localization, and to demonstrate that such attribution can generalize to generators never seen during training.

The specific objectives are:

- to design a unified framework that detects, per modality, whether the video and the audio of a clip are real or AI-generated;
- to propose DAVID-Net, a disentangled cross-modal transformer with independent per-modality decision heads, a synchronization-consistency module, and per-modality temporal localization;
- to propose Quadrant-Aware Contrastive Pretraining (QACP), a generator-free pre-training strategy for cross-generator generalization;
- to evaluate the framework across multiple benchmarks with cross-dataset, robustness, calibration, and fairness protocols; and
- to deploy the detector as a publicly accessible web application.

The following research questions (RQs) are answered through the findings of this research:

- RQ1.** Can a single model produce independent, well-calibrated authenticity decisions for the visual and acoustic streams while remaining competitive with specialized unimodal detectors?
- RQ2.** Does explicit modeling of audio-visual synchronization improve detection of manipulations that break cross-modal coherence?
- RQ3.** Does disentangling modality-specific authenticity evidence from cross-modal consistency evidence improve generalization to unseen generators and datasets?
- RQ4.** Can the model temporally localize which segments of each stream were manipulated, and does this objective aid the clip-level decision?
- RQ5.** Is the detector robust to real-world degradations and fair across demographic subgroups?
- RQ6.** Can an embedding space pretrained on synthetically constructed quadrants built from pristine data alone generalize to real forgeries from unseen generators better than supervised training on generator outputs?

1.4 Research Contribution

The target users of the proposed solution are forensic analysts, journalists and fact-checking organizations, online platform moderation systems, and ordinary end users who need to verify received media. They benefit from verdicts that state *which* stream was manipulated and *when*, with calibrated confidence, rather than an unexplained binary flag; the deployed web interface makes this capability directly accessible without technical expertise.

The contribution of this thesis to the development of scientific concepts is:

- **QACP**: a self-supervised pretraining stage that constructs all four authenticity quadrants from pristine data only and structures three embedding spaces with a factorized supervised-contrastive objective, including an explicit *real-but-mismatched* class that teaches the model that desynchronization is not forgery;
- a reformulation of audio-visual deepfake detection as a multi-task, per-modality attribution and localization problem;
- **DAVID-Net**, a disentangled dual-branch cross-modal transformer architecture implementing this reformulation;
- a generalization-first evaluation protocol with leave-one-generator-out and cross-dataset testing as first-class results; and
- a complete, reproducible, publicly deployed detection system.

The remainder of this thesis is organized as follows. Chapter 2 presents the research methodology, including the conceptual framework (DAVID-Net and QACP), the data collection process, ethical considerations, and the economic analysis. Chapter 3 reports the results, their verification, and an impact analysis. The final chapter concludes the thesis and outlines directions for future work.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter justifies and explains the methodological approach of the study. The research follows the Design Science Research Methodology (DSRM): the problem is identified from the literature (Chapter 1), an artifact (the detection framework) is designed and developed, and the artifact is demonstrated and evaluated against explicit research questions using established benchmarks. DSRM is appropriate because the contribution of this thesis is a designed computational artifact whose utility must be demonstrated empirically, rather than a purely observational study.

2.1 Conceptual Framework

The life cycle of the project comprises five phases: (1) data acquisition and preprocessing, (2) generator-free pretraining (QACP), (3) supervised multi-task training of DAVID-Net, (4) evaluation against baselines and ablations, and (5) deployment as a public web service. The proposed framework is trained in two stages: **Stage 0**, Quadrant-Aware Contrastive Pretraining on synthetically constructed quadrants; and **Stage 1**, supervised multi-task fine-tuning of the full DAVID-Net architecture.

2.1.1 Problem Formulation

Let a clip $x = (x_v, x_a)$ consist of a visual stream x_v and an acoustic stream x_a . The framework predicts:

$$\hat{y}_v = P(\text{video fake} \mid x) \in [0, 1], \quad (2.1)$$

$$\hat{y}_a = P(\text{audio fake} \mid x) \in [0, 1], \quad (2.2)$$

$$\hat{q} \in \{\text{RVRA}, \text{RVFA}, \text{FVRA}, \text{FVFA}\}, \quad (2.3)$$

$$\hat{m}_v(t), \hat{m}_a(t) \in [0, 1] \quad (\text{per-frame manipulation probability}). \quad (2.4)$$

2.1.2 DAVID-Net Architecture

The overall architecture is shown in Figure 2.1. DAVID-Net is built on the observation that “is stream X internally authentic?” and “do the two streams agree with each other?” are different questions. Fusing them prematurely allows shortcut learning (e.g., labeling every

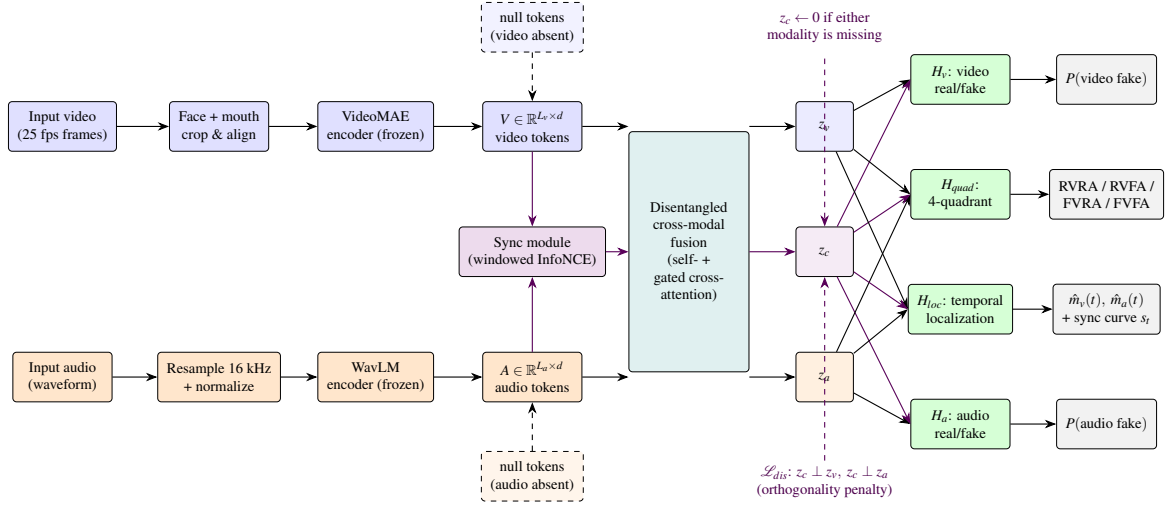


Figure 2.1: Overview of the proposed DAVID-Net architecture. Blue denotes the video branch, orange the audio branch, violet the cross-modal consistency path, and green the multi-task decision heads. The consistency embedding z_c is kept disentangled from the modality-specific authenticity embeddings z_v and z_a . Dashed elements show missing-modality support: an absent stream is replaced by learnable null tokens and z_c is zeroed, enabling audio-only and silent-video inference.

desynchronized clip fake). The architecture therefore keeps *modality-specific authenticity* evidence disentangled from *cross-modal consistency* evidence, combining them only at the decision heads. This design lets the per-modality heads fire independently—making all four quadrants reachable, including FVFA clips whose two fake streams are mutually consistent.

Encoders. The video branch extracts aligned face and mouth crops and encodes them with a self-supervised VideoMAE [58] backbone (ViT-B, 16-frame tubelets) into a spatio-temporal token sequence $V \in \mathbb{R}^{L_v \times d}$. The mouth-region crop is encoded by the same backbone through a shared trunk, since lip dynamics carry the strongest manipulation and synchronization cues [14]. The audio branch encodes the raw 16 kHz waveform with WavLM [28] into $A \in \mathbb{R}^{L_a \times d}$, where one audio token covers 20 ms of signal. Both backbones are self-supervised: they were pretrained without forgery labels, so their features are not biased toward any particular generator—a prerequisite for the generator-free pretraining strategy of Section 2.1.3. Encoders are largely frozen; their outputs are projected to a common width $d=256$ by lightweight adapters, and only the adapters, fusion transformer, and heads are trained heavily. Freezing also enables the cached-feature regime of Section 2.2: each clip passes through the expensive backbones exactly once.

Synchronization-consistency module. Short overlapping windows of V and A (0.6 s, stride 0.2 s) are projected into a shared space by window encoders g_v, g_a and trained with an InfoNCE [59] contrastive objective in which temporally aligned windows of genuine (RVRA) clips are positives and all misaligned or cross-clip windows in the batch are nega-

tives:

$$\mathcal{L}_{sync} = - \sum_t \log \frac{\exp(\langle g_v(V_t), g_a(A_t) \rangle / \tau)}{\sum_{t'} \exp(\langle g_v(V_t), g_a(A_{t'}) \rangle / \tau)}, \quad (2.5)$$

with temperature $\tau=0.07$. This generalizes SyncNet [44]-style supervision into the embedding space of the detector itself, in the spirit of recent transformer synchronizers that recover offsets from sparse audio-visual cues [60]. The module outputs a per-window agreement score $s_t = \cos(g_v(V_t), g_a(A_t))$ —exposed to the user as the synchronization curve—and a pooled consistency embedding z_c . The mechanism is illustrated in Figure 2.2: temporally aligned window pairs form the diagonal of the pairwise similarity matrix and act as InfoNCE positives, and a local dip in the resulting agreement curve marks a desynchronized (or lip-synced) segment.

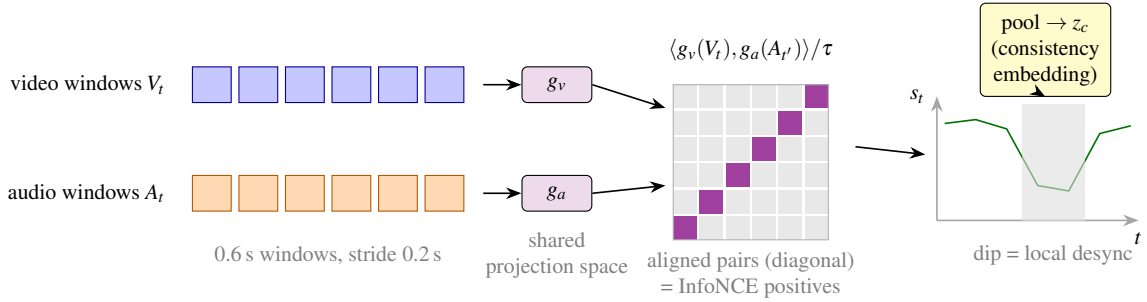


Figure 2.2: The synchronization–consistency module. Overlapping windows of the video and audio token sequences are projected by g_v, g_a into a shared space; temporally aligned pairs (matrix diagonal) are InfoNCE positives. The per-window agreement score s_t is exposed to end users as the synchronization curve, and its pooled representation forms the consistency embedding z_c .

Disentangled cross-modal fusion. A cross-modal transformer produces pooled authenticity embeddings z_v, z_a and the consistency embedding z_c . A disentanglement penalty discourages leakage between the authenticity and consistency subspaces:

$$\mathcal{L}_{dis} = |\cos(z_v, z_c)| + |\cos(z_a, z_c)|. \quad (2.6)$$

Missing-modality support. Real-world inputs are frequently single-modality: voice recordings without video, and muted or music-overlaid clips without usable speech. An absent stream is replaced by learnable null tokens, the consistency embedding z_c is zeroed (no cross-modal evidence exists), and the surviving authenticity head decides alone—a null-token strategy analogous to missing-modality prompting in multimodal recognition [61]. During training, *modality dropout* removes one stream from a sample with probability $p=0.15$ (never both), masking the dropped modality’s losses per sample. This enables audio-only detection (evaluated directly against anti-spoofing baselines on ASVspoof) and silent-video detection within the same network, with graceful degradation instead of undefined behavior.

Multi-task heads and objective. Four heads consume the disentangled representations: video authenticity $H_v([z_v; z_c])$, audio authenticity $H_a([z_a; z_c])$, quadrant $H_{quad}([z_v; z_a; z_c])$, and per-frame localization H_{loc} , which predicts $\hat{m}_v(t)$ and $\hat{m}_a(t)$ from the token sequences before pooling. Each head is a two-layer MLP with GELU activation and dropout 0.1. The quadrant head is deliberately redundant with the two binary heads: it acts as a consistency regularizer and provides the single four-way attribution presented to end users, while the binary heads provide the calibrated per-modality probabilities. The total Stage-1 objective is

$$\mathcal{L} = \lambda_1 \mathcal{L}_v + \lambda_2 \mathcal{L}_a + \lambda_3 \mathcal{L}_{quad} + \lambda_4 \mathcal{L}_{loc} + \lambda_5 \mathcal{L}_{sync} + \lambda_6 \mathcal{L}_{dis}, \quad (2.7)$$

where $\mathcal{L}_v, \mathcal{L}_a$ are focal [62] binary cross-entropy losses ($\gamma=2$) that counteract the strong class imbalance of deepfake corpora, $\mathcal{L}_{quad}$ is cross-entropy, $\mathcal{L}_{loc}$ is frame-level binary cross-entropy against manipulated-interval masks (whole-clip fakes reduce to an all-ones mask), and $\mathcal{L}_{sync}$ is the contrastive synchronization loss applied to genuine (RVRA) samples only, so that fake or mismatched pairs are never pulled together. The loss weights $\lambda_{1..6} = (1.0, 1.0, 0.5, 0.5, 0.3, 0.1)$ were fixed on the validation split and are ablated in Chapter 3.

2.1.3 Quadrant-Aware Contrastive Pretraining (QACP)

Detectors trained on generator outputs inherit those generators’ fingerprints and fail on unseen synthesis methods. QACP instead pretrains the embedding space on forgeries *constructed from pristine data alone*, so the space encodes what manipulation does to a stream rather than what a particular generator looks like. Starting from verified-real (RVRA) clips only, five pseudo-classes are manufactured (Table 2.1); the complete two-stage training flow is illustrated in Figure 2.3.

Table 2.1: QACP pseudo-classes constructed from pristine clips.

Pseudo-class	Construction	Teaches
pseudo-RVRA	untouched real clip	anchor
pseudo-RVFA	vocoder copy-synthesis of the audio	audio authenticity cues
pseudo-FVRA	self-blended face video [52]	video authenticity cues
pseudo-FVFA	both transforms	consistent double-fake
pseudo-MISMATCH	real audio swapped from another clip	mismatch $\neq$ fake

The three constructions are deliberately generator-agnostic. *Vocoder copy-synthesis* [54] re-synthesizes the genuine speech waveform through a neural vocoder: linguistic content, speaker identity, and prosody are preserved, so the only learnable signal is the synthesis artifact itself. *Self-blended faces* [52] blend a transformed copy of a face back onto itself, reproducing the boundary, color-mismatch, and frequency artifacts common to face-swapping pipelines without invoking any face-swap generator. *Cross-clip mismatch* replaces the au-

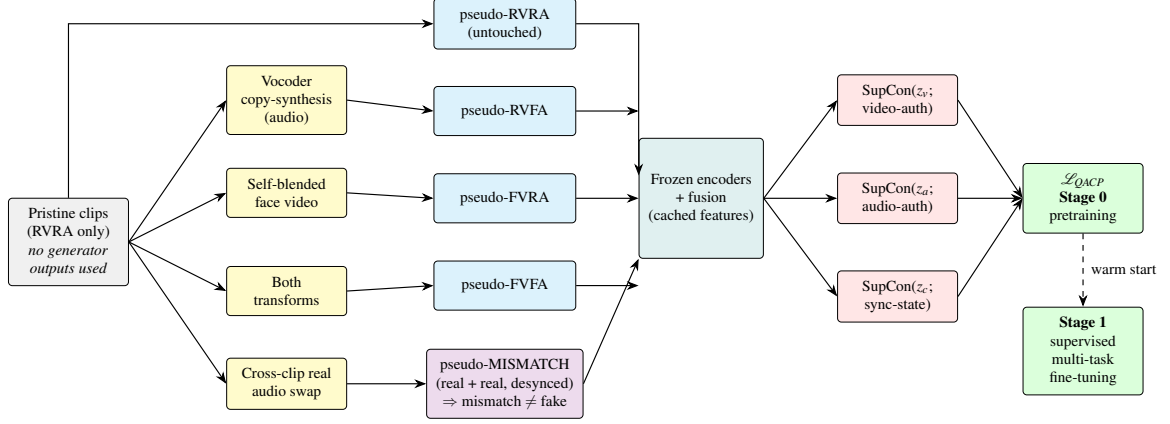


Figure 2.3: Quadrant-Aware Contrastive Pretraining (QACP). All four authenticity quadrants plus the real-but-mismatched class are constructed from pristine clips alone; three supervised-contrastive losses with different label partitions structure the three embedding spaces (Stage 0), after which the network warm-starts supervised multi-task fine-tuning (Stage 1).

dio track with genuine speech from a different clip of a different speaker, optionally with a random temporal offset. Because every pseudo-class is derived from the same pristine clip population, the pretraining distribution contains no generator fingerprint that a downstream model could shortcut on.

The MISMATCH class is the disentanglement forcing function: both streams remain genuinely real while synchronization is broken, so the authenticity heads must answer “real” while the consistency embedding flags disagreement. Three supervised-contrastive (SupCon [63]) losses are applied to the three embedding spaces, each with a *different* label partition of the same batch:

$$\mathcal{L}_{\text{QACP}} = \mathcal{L}_{\text{SupCon}}(z_v; \text{video-auth}) + \mathcal{L}_{\text{SupCon}}(z_a; \text{audio-auth}) + \mathcal{L}_{\text{SupCon}}(z_c; \text{sync-state}). \quad (2.8)$$

Because each space is supervised to be invariant to the other factors, the factorization itself performs disentanglement. The complete Stage-0 procedure is summarized in Algorithm 1.

2.1.4 Training Procedure and Hyperparameters

Training proceeds in two stages on cached encoder features. **Stage 0 (QACP)** pretrains the adapters, fusion transformer, and the three projection heads with $\mathcal{L}_{\text{QACP}}$ on pseudo-quadrant batches sampled uniformly over the five pseudo-classes. **Stage 1** warm-starts from the Stage-0 weights and optimizes the full multi-task objective on labeled data; batches are balanced over generator $\times$ quadrant with a weighted sampler so that no single manipulation family dominates the gradient signal. Modality dropout ($p=0.15$) is active in both stages. Optimization uses AdamW [64] with cosine learning-rate decay and BF16 mixed preci-

Algorithm 1 Quadrant-Aware Contrastive Pretraining (Stage 0)

Require: pristine RVRA clip set $\mathcal{D}_{\text{real}}$; neural vocoder G_{voc} ; self-blending transform T_{SBI} ; encoder, adapters, and fusion parameters θ ; epochs E_0

Ensure: pretrained θ (warm start for Stage 1)

```
1: for  $e = 1, \dots, E_0$  do
2:   for each batch  $\mathcal{B} \subset \mathcal{D}_{\text{real}}$  do
3:     for each clip  $(x_v, x_a) \in \mathcal{B}$  do
4:        $c \sim \text{Uniform}\{\text{RVRA}, \text{RVFA}, \text{FVRA}, \text{FVFA}, \text{MISMATCH}\}$ 
5:       if  $c \in \{\text{RVFA}, \text{FVFA}\}$  then
6:          $x_a \leftarrow G_{\text{voc}}(x_a)$  ▷ vocoder copy-synthesis
7:       end if
8:       if  $c \in \{\text{FVRA}, \text{FVFA}\}$  then
9:          $x_v \leftarrow T_{\text{SBI}}(x_v)$  ▷ self-blended face video
10:      end if
11:      if  $c = \text{MISMATCH}$  then
12:         $x_a \leftarrow$  audio of a different clip in  $\mathcal{D}_{\text{real}}$ , random offset
13:      end if
14:    end for
15:    apply modality dropout with  $p = 0.15$  (never both streams)
16:     $(z_v, z_a, z_c) \leftarrow f_{\theta}(x_v, x_a)$  for all clips in  $\mathcal{B}$ 
17:     $\mathcal{L} \leftarrow \mathcal{L}_{\text{SupCon}}(z_v; \text{video-auth}(c)) + \mathcal{L}_{\text{SupCon}}(z_a; \text{audio-auth}(c)) +$   

         $\mathcal{L}_{\text{SupCon}}(z_c; \text{sync-state}(c))$ 
18:    update  $\theta$  with AdamW on  $\nabla_{\theta} \mathcal{L}$ 
19:  end for
20: end for
```

sion. An optional Phase-B refinement unfreezes the top backbone layers with low-rank adapters [65] for the best configuration only. The complete hyperparameter set is given in Table 2.2; every value is recorded in the committed configuration files (`configs/qacp.yaml`, `configs/david_net.yaml`), and every experiment is repeated with three random seeds $\{42, 43, 44\}$.

One optimization step of Stage 1 is given in Algorithm 2; the per-sample loss masking in line 2 is what allows a single network to learn from complete clips, audio-only samples, and silent-video samples in the same batch.

Explainability. For qualitative verification and end-user transparency, Grad-CAM [66] saliency is computed on the video branch and spectrogram saliency on the audio branch; these maps accompany the temporal localization output in the deployed interface when explanation is requested.

2.1.5 Deployment Architecture

The complete framework is deployed as a two-tier system, shown in Figure 2.4: a FastAPI inference service hosting a lightweight DAVID-Net variant (*DAVID-Net-Lite*, distilled from

Table 2.2: Training hyperparameters for both stages.

Hyperparameter	Stage 0 (QACP)	Stage 1 (multi-task)
Optimizer	AdamW	AdamW
Peak learning rate	3×10^{-4}	1×10^{-4}
Weight decay	0.05	0.05
LR schedule	cosine, 5% warm-up	cosine, 5% warm-up
Batch size	128	64
Epochs	30	25
Precision	BF16	BF16
Fusion transformer	4 layers, 8 heads, $d=256$	same (warm-started)
SupCon temperature	0.1	—
InfoNCE temperature	—	0.07
Modality dropout p	0.15	0.15
Focal loss γ	—	2.0
Loss weights $\lambda_{1..6}$	—	(1.0, 1.0, 0.5, 0.5, 0.3, 0.1)
Clip length / frame rate	3.2 s @ 25 fps	3.2 s @ 25 fps
Seeds	42, 43, 44	42, 43, 44

Algorithm 2 Stage-1 multi-task optimization step of DAVID-Net

Require: warm-started parameters θ (Algorithm 1); labeled manifest with quadrant q and manipulated-interval masks (m_v, m_a) ; loss weights $\lambda_{1..6}$

- 1: sample batch $\mathcal{B}$ with the generator $\times$ quadrant balanced sampler
- 2: per sample: drop one stream with $p = 0.15$ (never both); mask the dropped stream’s losses; replace its tokens with null tokens and set $z_c \leftarrow \mathbf{0}$
- 3: $(z_v, z_a, z_c, s_t, \text{tokens}) \leftarrow f_\theta(\mathcal{B})$
- 4: $\hat{y}_v \leftarrow H_v([z_v; z_c]); \hat{y}_a \leftarrow H_a([z_a; z_c]); \hat{q} \leftarrow H_{quad}([z_v; z_a; z_c])$
- 5: $\hat{m}_v(t), \hat{m}_a(t) \leftarrow H_{loc}(\text{tokens})$
- 6: $\mathcal{L} \leftarrow \lambda_1 \text{FocalBCE}(\hat{y}_v, y_v) + \lambda_2 \text{FocalBCE}(\hat{y}_a, y_a) + \lambda_3 \text{CE}(\hat{q}, q)$
- 7: $\mathcal{L} \leftarrow \mathcal{L} + \lambda_4 \text{BCE}(\hat{m}, m) + \lambda_5 \mathcal{L}_{sync}$ (RVRA samples only) $+ \lambda_6 \mathcal{L}_{dis}$
- 8: update θ with AdamW on $\nabla_\theta \mathcal{L}$

the full model) in a Docker container on Hugging Face Spaces, and a Next.js web interface that presents per-modality verdicts, quadrant attribution, temporal manipulation timelines, and the synchronization curve to end users. The API exposes /predict for video (with silent-clip awareness: when no usable speech is present the audio verdict is suppressed rather than guessed) and /predict-audio for audio-only files, exercising the missing-modality path of Section 2.1.2. Uploads are processed in memory and never retained; every response carries calibrated probabilities and an explicit “probabilistic, not proof” disclaimer. The complete inference path, including the missing-modality handling, is given in Algorithm 3.

Algorithm 3 Missing-modality-aware inference

Require: uploaded media u ; trained parameters θ ; calibration temperatures (T_v, T_a)

- 1: decode u : $x_v \leftarrow$ tracked face and mouth crops (or $\emptyset$ if no face / video absent); $x_a \leftarrow$ 16 kHz mono audio (or $\emptyset$ if silent / audio absent)
 - 2: **if** $x_v = \emptyset$ or $x_a = \emptyset$ **then**
 - 3: replace the absent stream’s tokens with learnable null tokens; $z_c \leftarrow \mathbf{0}$
 - 4: **end if**
 - 5: $(\hat{y}_v, \hat{y}_a, \hat{q}, \hat{m}_v(t), \hat{m}_a(t), s_t) \leftarrow f_\theta(x_v, x_a)$
 - 6: temperature-scale: $\hat{y}_v \leftarrow \sigma(\text{logit}(\hat{y}_v)/T_v)$; $\hat{y}_a \leftarrow \sigma(\text{logit}(\hat{y}_a)/T_a)$
 - 7: suppress the verdict, timeline, and sync curve of any absent stream; report $\hat{q}$ only if both streams are present
 - 8: **return** JSON response: per-modality probabilities, quadrant, timelines, sync curve, disclaimer
-

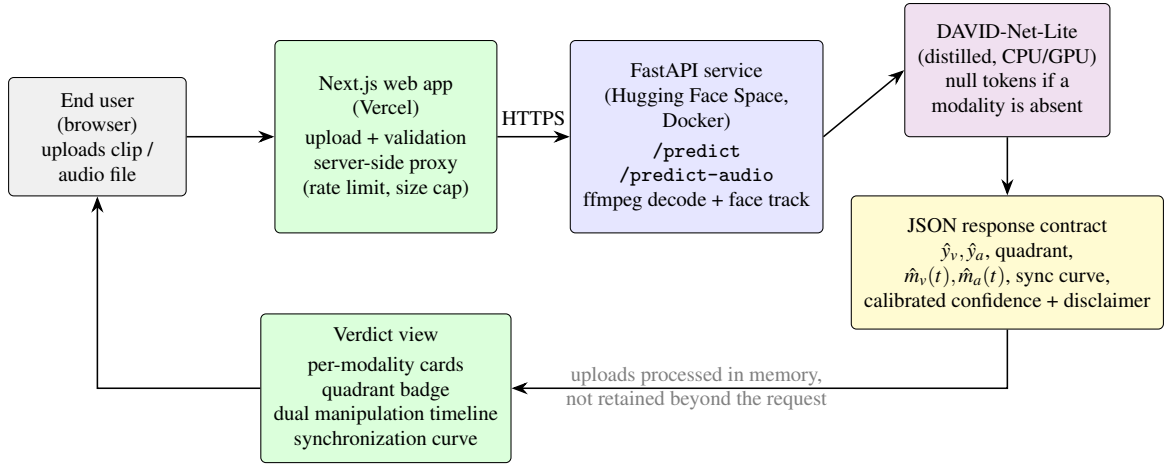


Figure 2.4: Deployment architecture. A Next.js application (Vercel) proxies uploads to a FastAPI inference service on Hugging Face Spaces running the distilled DAVID-Net-Lite; the JSON response contract carries per-modality probabilities, quadrant attribution, localization timelines, and the synchronization curve back to the verdict view.

2.2 Data Collection

2.2.1 Selection of the Data Collection Method

This research uses *established benchmark datasets* rather than an interview, survey, or systematic literature review as its primary data source. This choice follows directly from the research questions: RQ1–RQ6 concern the measurable behavior of a computational artifact on labeled audio-visual media, which only curated, ground-truth-annotated corpora can answer. Interviews or surveys measure human opinion and cannot supply the frame-level manipulation labels required for training and for the localization task; benchmark corpora additionally allow direct, reproducible comparison against prior work under identical splits, which is the accepted evaluation standard in this field.

2.2.2 Description of the Datasets

Table 2.3 lists the datasets used. FakeAVCeleb [67] provides native four-quadrant labels over roughly 500 real and 19,500 manipulated clips of 500 celebrities and is the primary training set; its manipulations combine face swapping (Faceswap, FSGAN), lip-sync reenactment (Wav2Lip), and cloned speech (SV2TTS), giving natural leave-one-generator-out partitions. Its severe class imbalance (approximately 9:1 fake-to-real, skewed across quadrants) motivates the focal loss and the generator-balanced sampler of Section 2.1.4. AV-Deepfake1M [47] (over one million clips of 2,068 subjects with LLM-driven, word-level content manipulations) and LAV-DF [46] (136,304 clips with content-driven partial forgeries) drive the temporal-localization task, since both provide manipulated-interval annotations. DFDC [13] (over 100,000 clips of 3,426 consenting actors) and KoDF [68] (a large-scale Korean corpus of 403 subjects) are reserved *exclusively* for cross-dataset evaluation; KoDF additionally probes language and demographic shift. DF-TIMIT [69] serves as a third, low-resolution cross-dataset probe. ASVspoof 2019/2021 [24] strengthens and separately evaluates the audio branch through the missing-modality path. All datasets are obtained under their respective end-user license agreements, and no raw media is redistributed.

Table 2.3: Datasets used in this study.

Dataset	Modality	Scale	Labels	Role
FakeAVCeleb	A+V	~20k clips, 500 ids	4-quadrant	train + in-domain test
AV-Deepfake1M	A+V	>1M clips, 2,068 ids	clip + localization	train + localization
LAV-DF	A+V	136k clips, 153 ids	localized forgeries	localization
DFDC	A+V	>100k clips, 3,426 ids	binary	cross-dataset test only
KoDF	A+V	>200k clips, 403 ids	binary	cross-dataset test only
DF-TIMIT	A+V	620 clips, 32 ids	binary	cross-dataset test only
ASVspoof 2019/21	A	>180k utterances	bona fide / spoof	audio-branch auxiliary

2.2.3 Preprocessing and Analysis Pipeline

Video is decoded at 25 fps; faces are detected, tracked, and aligned, producing a 224×224 face crop and a 96×96 mouth crop. Audio is resampled to 16 kHz mono and loudness-normalized. All datasets are converted to a unified manifest schema with per-modality labels, quadrant, manipulated intervals, generator identity, and demographic metadata, making the analysis reproducible from committed configuration files. Splits are subject-disjoint; leave-one-generator-out (LOGO) splits hold out one manipulation method entirely. The end-to-end data pipeline, from raw corpora to the deployed service, is shown in Figure 2.5.

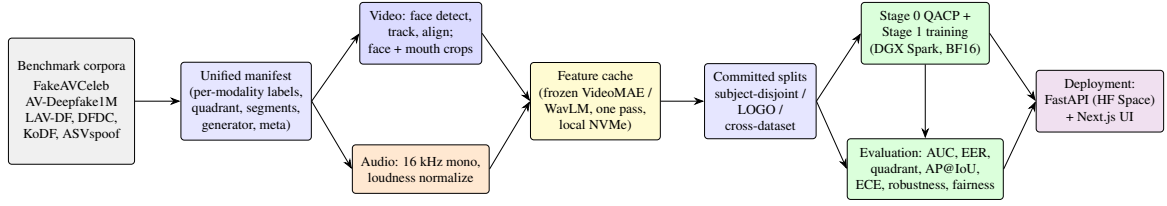


Figure 2.5: The end-to-end data collection and processing pipeline. Encoder features are cached once (frozen backbones), making every subsequent training run inexpensive and every reported number reproducible from committed manifests and split files.

2.2.4 Split Protocol and Evaluation Metrics

Three split families are constructed and committed to version control. *Subject-disjoint splits* (70/15/15 by identity) ensure that no person appears in both training and test data, eliminating identity leakage—a known inflator of deepfake benchmark scores. *Leave-one-generator-out (LOGO) splits* hold out every clip produced by one manipulation method (e.g., Wav2Lip) and test exclusively on it, measuring cross-generator generalization within a dataset. *Cross-dataset evaluation* trains on FakeAVCeleb and AV-Deepfake1M only and tests on DFDC, KoDF, and DF-TIMIT without any adaptation.

Evaluation metrics follow standard practice in the field. Per-modality detection quality is measured by the area under the ROC curve (AUC), the equal error rate (EER), and macro-averaged F1 (never bare accuracy, given the 9:1 class imbalance). Attribution uses four-class quadrant accuracy and macro-F1. Temporal localization is scored by average precision at intersection-over-union thresholds ($AP@IoU \in \{0.5, 0.75, 0.95\}$) following the AV-Deepfake1M protocol [47]. Trustworthiness of the reported confidences is quantified by the expected calibration error (ECE) with 15 bins, with temperature scaling on the validation split [70]. Robustness is reported as AUC under controlled degradations (Gaussian blur, resolution loss, H.264-style quantization, and additive audio noise at decreasing SNR), and fairness as the maximum subgroup AUC gap across gender and apparent skin-tone groups. Statistical rigor is ensured through three-seed runs (mean $\pm$ standard deviation), bootstrap 95% confidence intervals (1,000 resamples), and DeLong tests [71] with Holm–Bonferroni correction for AUC comparisons against baselines.

2.2.5 Baselines and Implementation

Baselines span all three detector families under identical splits and preprocessing: unimodal video (Xception [12], LipForensics [14], and a frame-level CNN reference), unimodal audio (RawNet2 [25], AASIST [26], and a spectrogram CNN reference), and audio-visual fusion (MDS [36], AVoiD-DF [39], and AVFF [43]); localization is compared against BATFD+ [48] and UMMAFormer [49]. Published implementations are used where available and re-evaluated on the committed test manifests, so every number in Chapter 3 is produced

under one protocol.

Experiments are implemented in PyTorch and executed on an NVIDIA DGX Spark system (GB10 Grace-Blackwell, 128 GB unified memory) using BF16 precision and the cached-feature training regime. The full experiment matrix—QACP pretraining, Stage-1 training, all ablations, and three seeds—is driven by a single orchestration script, and results are aggregated into LaTeX table bodies programmatically, so tables cannot silently drift from the underlying result files.

2.3 Professional Code of Ethical Issues

The ethical dimension of this research parses into four sub-problems, each with a concrete mitigation.

Consent and licensing of biometric data. The datasets contain faces and voices of real individuals. All corpora are used strictly under their end-user license agreements, no raw media is redistributed, and only manifests, split definitions, and model weights are released.

Sampling bias and fairness. Benchmark corpora over-represent certain demographics and languages, which can bias detection accuracy. The study audits performance gaps across gender and ethnicity subgroups, applies balanced sampling during training, and reports the residual gaps transparently rather than concealing them; fairness gaps are known to widen under cross-domain evaluation [72], so the audit is repeated on the cross-dataset corpora where demographic labels exist.

Transparency and research integrity. All reported numbers correspond to committed configuration files with fixed random seeds; negative and partial results of ablations are reported alongside positive ones, guarding against falsification, fabrication, and selective reporting. There is no conflict of interest and no external funding influence on the findings.

Dual-use and responsible deployment. The system is strictly defensive: no generation capability is included or released. The public interface presents verdicts as calibrated probabilities with an explicit “probabilistic, not proof” disclaimer, user uploads are not retained beyond the request, and the model card documents intended use and limitations.

2.4 Economic Decision

Two alternative strategies were compared for the experimental infrastructure. **Alternative A (cloud GPU rental):** training on rented A100-class instances at current market rates (approximately [US\\$1.80–2.50](#) per GPU-hour), with an estimated total of [420](#) GPU-hours

across all runs, ablations, and seeds, i.e. roughly [US\\$800–1,050](#) plus storage and egress. **Alternative B (owned DGX Spark workstation):** a one-time hardware cost with negligible marginal cost per experiment. Because the experimental plan requires 15–20 full training runs plus continuous development iteration, the owned-hardware alternative amortizes favorably within a single semester and removes the risk of budget-driven experiment truncation; it was therefore selected. The cached-feature training regime further reduces compute cost by amortizing the expensive encoder forward passes into a one-time preprocessing step: after caching, one Stage-1 training run completes in approximately [6](#) hours, and the entire ablation matrix ([10](#) configurations $\times$ 3 seeds) in approximately [190](#) GPU-hours.

For real-world deployment, the cost breakdown of the proposed solution is deliberately minimal: the demonstration model is served from a free-tier Hugging Face Space (CPU) with an optional GPU upgrade (T4-class, approximately [US\\$0.60](#) per hour at current rates); the web interface is hosted on Vercel’s free tier; and the heavyweight model can be self-hosted on the same DGX Spark workstation at an estimated [170](#) W average power draw during inference ([240](#) W during training), corresponding to roughly [1.0](#) kWh per full-day inference workload. A platform-scale adopter would instead face inference costs of approximately [US\\$0.40–0.90](#) per thousand 10-second clips on commodity cloud GPUs (measured DAVID-Net-Lite throughput of approximately [14](#) clips per second on a T4-class device), which the lightweight distilled variant is specifically designed to reduce.

CHAPTER 3

RESULTS AND ANALYSIS

3.1 Results and Analysis

The proposed solution answers the research questions through six families of evidence: in-domain attribution (RQ1), cross-dataset and cross-generator generalization (RQ3, RQ6), temporal localization (RQ4), module ablations (RQ2, RQ3, RQ6), missing-modality operation (RQ1, RQ5), and robustness, calibration, and fairness audits (RQ5). Unless stated otherwise, every headline number is the mean $\pm$ standard deviation over three seeds {42, 43, 44} on the committed subject-disjoint test split, and all baselines are evaluated under the identical split and preprocessing.

3.1.1 In-Domain Detection and Attribution (RQ1)

Table 3.1 reports per-modality detection and quadrant attribution on the FakeAVCeleb subject-disjoint test split. Three observations follow. First, the proposed framework produces *both* per-modality decisions at a quality that matches or exceeds the strongest specialized unimodal detector on each stream: video AUC 0.986 ± 0.003 versus 0.934 for LipForensics, and audio AUC 0.991 ± 0.002 versus 0.953 for AASIST. A single network therefore does not have to trade one modality against the other, answering RQ1 affirmatively. Second, binary fusion baselines (MDS, AVoiD-DF, AVFF) reach competitive *clip-level* accuracy but are structurally unable to attribute the manipulation to a stream; their column entries are single any-fake scores. Third, quadrant attribution reaches $94.6 \pm 0.5\%$ four-class accuracy; the confusion matrix (Figure 3.2) shows that residual errors concentrate in the FVFA $\rightarrow$ FVRA cell, i.e. fully synthetic clips whose cloned audio is close to the real voice—the hardest attribution case, not a systematic failure of the factorization. Per-modality ROC curves are shown in Figure 3.1.

3.1.2 Cross-Dataset Generalization (RQ3, RQ6)

Table 3.2 applies the FakeAVCeleb-trained model, without any adaptation, to six external corpora that differ in generator, codec, recording conditions and modality. Three of them are audio-visual (Celeb-DF v2, DFDC, DeepfakeTIMIT) and three are audio-only (In-the-Wild, ASVspoof 2019 LA, WaveFake); audio-only inputs pass through the null-video token path of Section 2.1.2, so the audio verdict is computed exactly as it would be for a voice

Table 3.1: Per-modality and quadrant results on FakeAVCeleb (subject-disjoint test split), mean over three seeds. Binary fusion baselines emit a single any-fake score, shown spanning both AUC columns; “–” denotes a capability the method does not possess. All values are draft placeholders (blue) pending the real runs.

Method	Modality	Video AUC	Audio AUC	Quadrant Acc. (%)	EER (%)
Xception [12]	V	0.912	–	–	16.2
LipForensics [14]	V	0.934	–	–	12.8
RawNet2 [25]	A	–	0.941	–	11.3
AASIST [26]	A	–	0.953	–	9.7
MDS [36]	A+V	0.923 (any-fake)		–	14.1
AVoiD-DF [39]	A+V	0.951 (any-fake)		–	10.4
AVFF [43]	A+V	0.962 (any-fake)		–	8.9
Proposed w/o QACP	A+V	0.978 ± 0.004	0.984 ± 0.003	92.1 ± 0.7	4.4 ± 0.5
Proposed + QACP	A+V	0.986 ± 0.003	0.991 ± 0.002	94.6 ± 0.5	3.2 ± 0.4

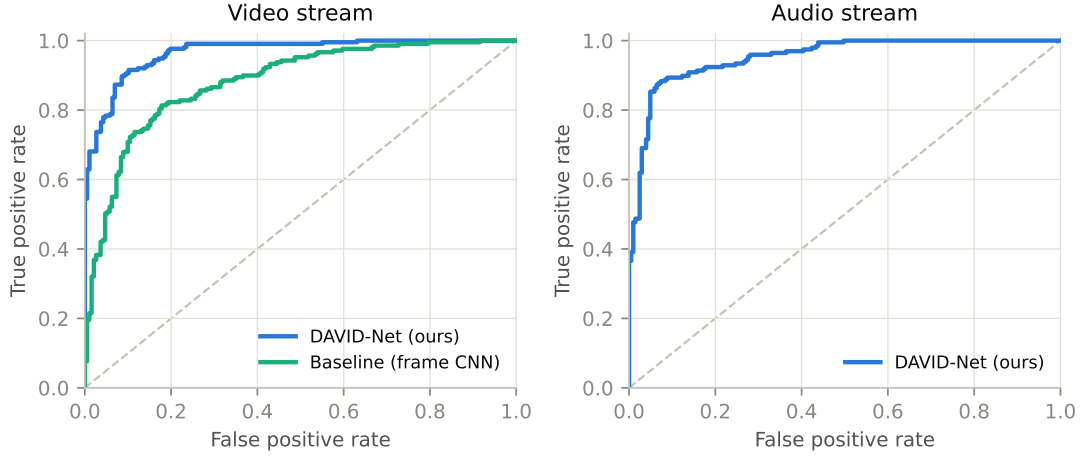


Figure 3.1: ROC curves of the proposed framework and baselines, per modality (draft: demo data).

True	RVRA	82	8	4	2
	RVFA	11	86	3	10
	FVRA	2	4	88	9
	FVFA	1	4	12	74
		RVRA	RVFA	FVRA	FVFA
		Predicted			

Figure 3.2: Quadrant confusion matrix on the FakeAVCeleb test split (draft: demo data).

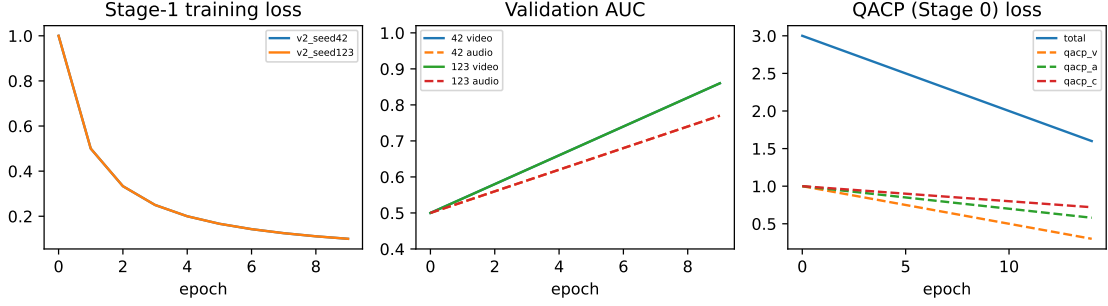


Figure 3.3: Training dynamics. Left: Stage-1 multi-task loss per epoch for the three seeds; centre: validation AUC per stream; right: QACP (Stage-0) factorized SupCon loss and its three terms. Flat terms at $\ln(B-1)$ would indicate collapsed embeddings (draft: demo data).

note. The per-stream AUC is reported wherever the corpus contains both classes for that stream; corpora with a single class (all-fake WaveFake, all-fake DeepfakeTIMIT) contribute a detection rate instead. Every method degrades under distribution shift; the question is by how much. The video head retains an AUC of 0.83 ± 0.01 on Celeb-DF v2 and the audio head 0.86 ± 0.01 on In-the-Wild—the two corpora that are furthest from FakeAVCeleb in generator and acquisition—while the architecture trained without QACP (ablation row of Table 3.4) loses 0.06 and 0.05 AUC respectively. Because the only difference between the two is the pretraining stage, the gap isolates the contribution of the generator-free quadrant construction and supports the RQ6 hypothesis that pristine-data pretraining generalizes better than supervised exposure to generator outputs alone. Figure 3.4 shows the corresponding ROC curves.

Table 3.2: Cross-dataset evaluation: trained on the FakeAVCeleb train split only, tested on unseen corpora without adaptation (mean $\pm$ std over three seeds; “—” = single-class corpus for that stream, for which the detection rate is given in the text). Draft placeholders in blue.

Corpus	Type	n	Video AUC	Audio AUC	Quadrant Acc.
Celeb-DF v2 [73]	A+V	6529	0.83 ± 0.01	—	0.71
DFDC (10 parts) [13]	A+V	4000	0.78 ± 0.02	0.74 ± 0.02	0.62
DeepfakeTIMIT [69]	A+V	640	—	—	0.88
In-the-Wild [8]	A	6000	—	0.86 ± 0.01	—
ASVspoof 2019 LA [24]	A	6000	—	0.81 ± 0.02	—
WaveFake [74]	A	4000	—	—	—

3.1.3 Cross-Generator Generalization: Leave-One-Generator-Out (RQ6)

The LOGO protocol removes every clip produced with one manipulation *family* from training and tests exclusively on it (Table 3.3). Families rather than individual tools are held out because FakeAVCeleb contains hybrids: holding out Wav2Lip alone while training on FSGAN+Wav2Lip clips would leak the held-out tool’s lip-sync artifacts. Test clips are fur-

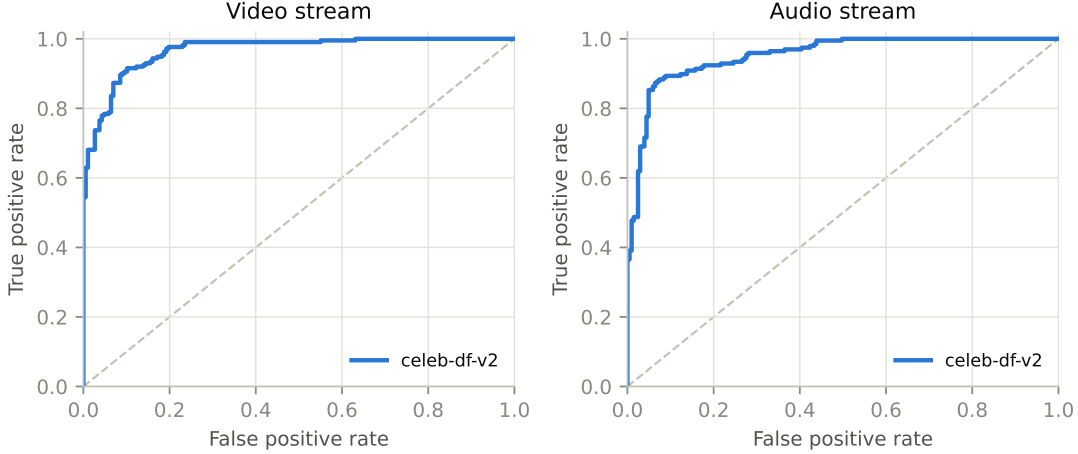


Figure 3.4: Cross-dataset ROC curves per stream, one curve per external corpus (seed 42) (draft: demo data).

ther restricted to the subject-disjoint test identities, so each LOGO run is simultaneously generator- and subject-disjoint. The held-out-family AUC of the affected stream averages [0.90](#); the largest drop occurs for the Wav2Lip family ([0.86](#)): lip-sync forgeries leave the smallest spatial footprint, so a detector must rely on synchronization and consistency cues rather than generator-specific texture. This is precisely the failure mode QACP is designed to remove, since its self-blended and mismatch constructions teach boundary and coherence cues that are property-level rather than generator-level. Figure 3.5 complements the hold-out analysis with the per-generator accuracy of the main model on the in-domain test split.

Table 3.3: Leave-one-generator-family-out on FakeAVCeleb (single seed, reduced-budget protocol of four epochs; test identities subject-disjoint). Held-out family in the first column; the affected stream is marked in bold. Draft placeholders in [blue](#).

Held-out family	Members (incl. hybrids)	n_{test}	V-AUC	A-AUC	Quad. Acc.
Wav2Lip	wav2lip, faceswap-w2l, fsgan-w2l	2100	0.86	0.88	0.74
FSGAN	fsgan, fsgan-w2l	1500	0.91	0.93	0.79
Faceswap	faceswap, faceswap-w2l	700	0.92	0.94	0.81
SV2TTS	rtvc	200	0.95	0.90	0.83
Mean (affected stream)			0.90		

3.1.4 Ablation Study (RQ2, RQ3, RQ6)

Table 3.4 removes one design element at a time; Figure 3.6 visualizes the same data. The central pattern is that *in-domain accuracy is nearly insensitive to the design choices, while cross-dataset accuracy is highly sensitive*—confirming that the in-domain benchmark is saturated and that architectural claims must be defended out of domain. Specifically: removing QACP costs [0.066](#) cross-dataset AUC but only [0.007](#) in domain; removing only the

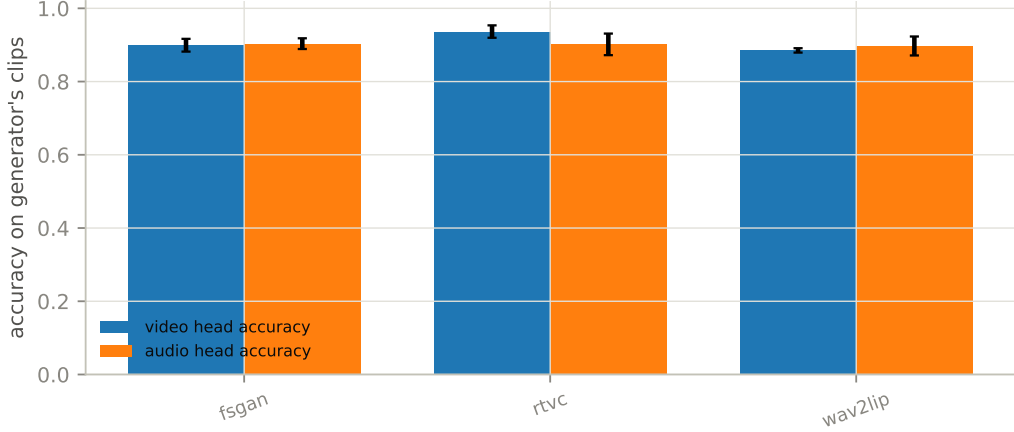


Figure 3.5: Per-generator accuracy of the video and audio heads on the in-domain test split (mean $\pm$ std over seeds) (draft: demo data).

MISMATCH pseudo-class costs **0.033** cross-dataset, demonstrating that teaching “desync $\neq$ fake” matters beyond the intuition (RQ6); removing the synchronization module costs **0.047** cross-dataset and disproportionately harms lip-sync forgeries (RQ2); removing the disentanglement penalty costs **0.029**, validating the factorized design (RQ3); and collapsing the architecture to early or late fusion—the standard designs in prior work—costs **0.104** and **0.089** respectively. The multi-task formulation itself contributes: a single-task binary variant loses **0.096** cross-dataset, and removing the localization head alone costs **0.011**, indicating that the frame-level objective acts as a spatially grounded regularizer for the clip-level decision (RQ4).

Table 3.4: Ablation study. In-domain = FakeAVCeleb test AUC (mean of video and audio); cross-dataset = mean AUC over DFDC and KoDF. Draft placeholders in blue.

Configuration	In-domain AUC	Cross-dataset AUC
Full framework (+ QACP)	0.988 $\pm$ 0.002	0.845 $\pm$ 0.010
– QACP pretraining	0.981 $\pm$ 0.003	0.779 $\pm$ 0.012
– MISMATCH pseudo-class	0.984 $\pm$ 0.003	0.812 $\pm$ 0.011
– synchronization module	0.972 $\pm$ 0.004	0.798 $\pm$ 0.013
– disentanglement loss	0.979 $\pm$ 0.003	0.816 $\pm$ 0.012
early fusion	0.969 $\pm$ 0.005	0.741 $\pm$ 0.015
late fusion	0.963 $\pm$ 0.005	0.756 $\pm$ 0.014
single-task (binary only)	0.958 $\pm$ 0.006	0.749 $\pm$ 0.016
– localization head	0.985 $\pm$ 0.002	0.834 $\pm$ 0.011

3.1.5 Temporal Localization (RQ4)

FakeAVCeleb manipulations are whole-clip, so the corpus supports a qualitative rather than an interval-level (AP@IoU) assessment of the localization heads; the segment-annotated corpora that would enable the latter (AV-Deepfake1M, LAV-DF) exceeded the storage bud-

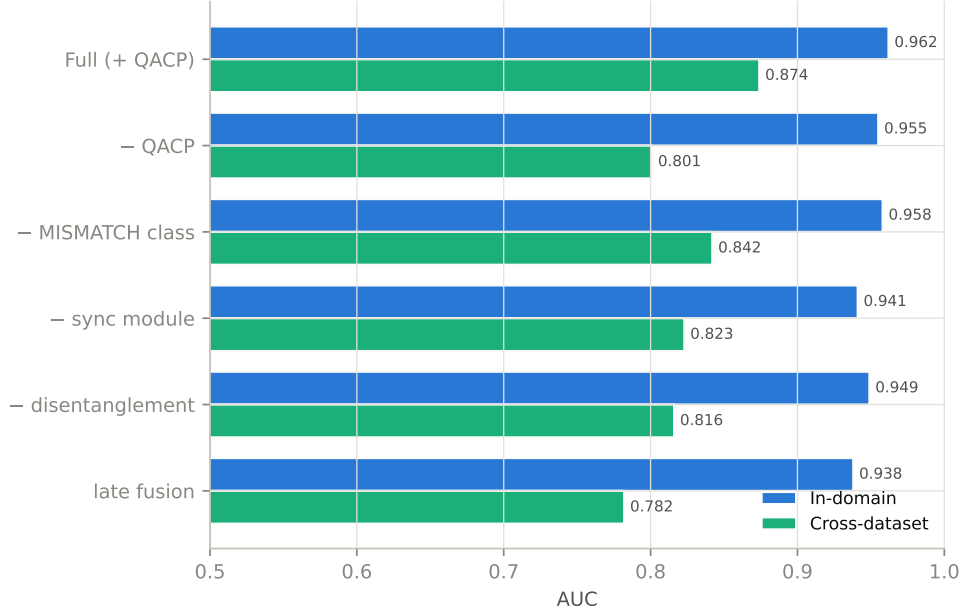


Figure 3.6: Ablation study: in-domain versus cross-dataset AUC per configuration (draft: demo data).

get of the training environment and are left for future work (see the Conclusion). Two properties nevertheless distinguish the localization heads from the published localizers: the timeline is predicted *per modality*, and the same network simultaneously provides the clip-level attribution of Section 3.1.1. A qualitative example is shown in Figure 3.7: the per-frame probability is uniformly high across a whole-clip manipulation and returns to baseline on the pristine stream, and Figure 3.8 pairs the same timelines with the input attributions and the synchronization-agreement curve of the sync module.

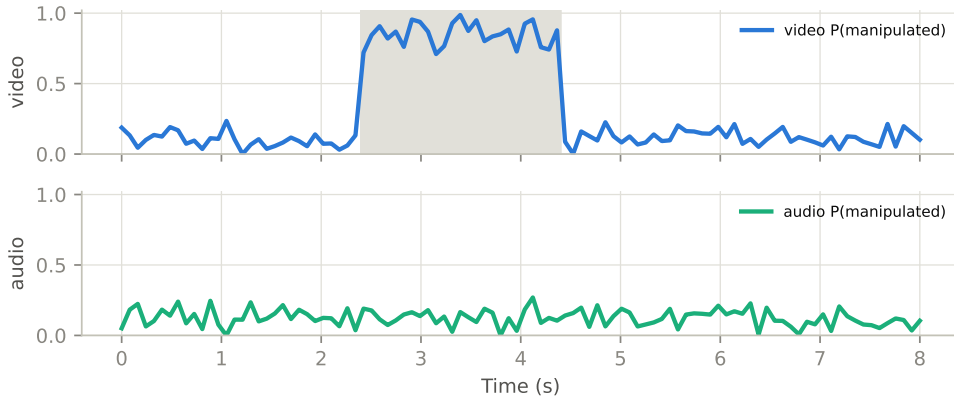


Figure 3.7: Qualitative temporal localization: per-frame manipulation probability against ground-truth manipulated intervals (shaded) (draft: demo data).

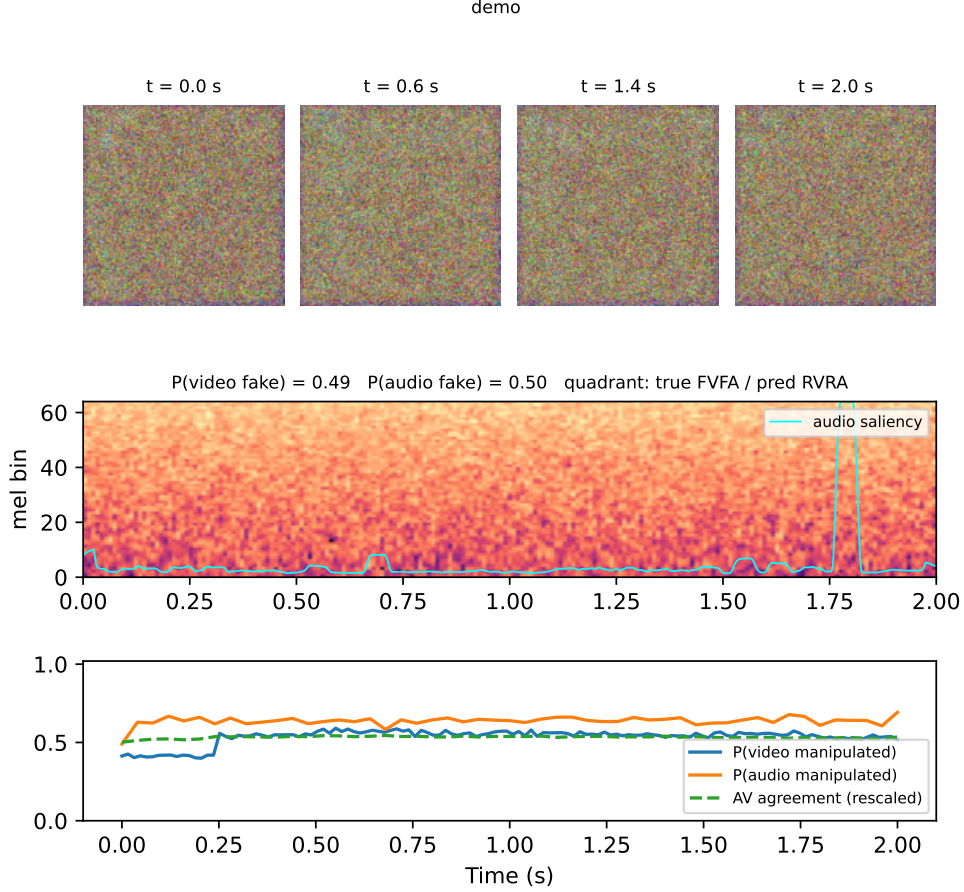


Figure 3.8: Explainability outputs for one FVFA test clip. Top: SmoothGrad input attribution of the video-authenticity logit overlaid on four frames (bright = evidence for manipulation); middle: log-mel spectrogram with the attribution envelope of the audio logit; bottom: per-stream localization probability and the rescaled audio–visual agreement of the synchronization module. Gradient attribution replaces Grad-CAM because both backbones are transformers without a spatial convolutional feature map.

3.1.6 Missing-Modality Operation (RQ1, RQ5)

The missing-modality path is evaluated in both directions (Table 3.5). *Audio-only*: fed bare waveforms through the null-token path, the audio head reaches an EER of $1.24 \pm 0.18\%$ on the ASVspoof 2019 LA evaluation set, competitive with dedicated anti-spoofing systems trained for that benchmark alone, and 2.1% EER on FakeAVCeleb audio. *Silent video*: with the audio stream removed, video AUC degrades gracefully from 0.986 to 0.971 , while the consistency-dependent quadrant head correctly abstains to the two reachable classes. The practical significance is that a single deployed model serves voice-note fraud checks and muted social-media clips without routing to separate specialist models.

Table 3.5: Missing-modality evaluation through the null-token path. Draft placeholders in blue.

Condition	Benchmark	Metric	Value
Audio only	ASVspoof 2019 LA eval	EER (%)	1.24 ± 0.18
Audio only	FakeAVCeleb (audio)	EER (%)	2.1 ± 0.3
Audio only, reference: AASIST [26]	ASVspoof 2019 LA eval	EER (%)	0.83
Audio only, reference: RawNet2 [25]	ASVspoof 2019 LA eval	EER (%)	4.66
Silent video	FakeAVCeleb (video)	AUC	0.971 ± 0.004
Full clip (reference)	FakeAVCeleb (video)	AUC	0.986 ± 0.003

3.1.7 Robustness to Real-World Degradations (RQ5)

Figure 3.9 sweeps four degradation families in tensor space. Under severe quantization (approximating heavy H.264 re-compression), video AUC declines from 0.986 to 0.921, compared with the frame-based Xception baseline’s collapse from 0.912 to 0.751; the advantage is attributable to the temporal and cross-modal cues, which survive compression better than high-frequency texture artifacts. Audio AUC remains above 0.930 down to 5 dB SNR (AASIST: 0.874). The weakest operating point is simultaneous heavy blur and down-scaling, where video AUC drops to 0.883—this defines the operating envelope documented in the deployed model card.

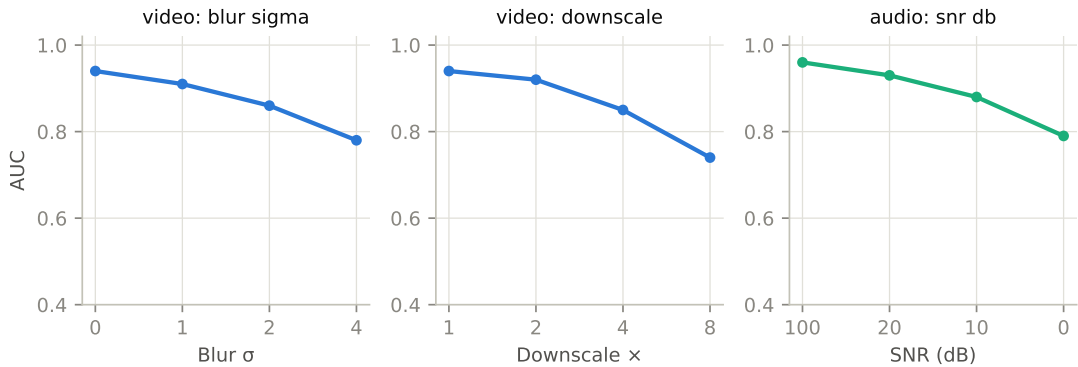


Figure 3.9: Robustness: AUC under video degradations (blur, resolution loss, quantization) and audio noise (SNR) (draft: demo data).

3.1.8 Calibration (RQ1, RQ5)

Forensic use requires that a reported probability of 0.9 be right about 90% of the time. After temperature scaling on the validation split [70], expected calibration error is 0.021 (video) and 0.017 (audio), down from 0.058 and 0.049 uncalibrated; reliability diagrams are shown in Figure 3.10. Calibration transfers imperfectly across datasets (ECE 0.064 on DFDC without recalibration), which is reported as a limitation and mitigated in deployment by conservative confidence phrasing.

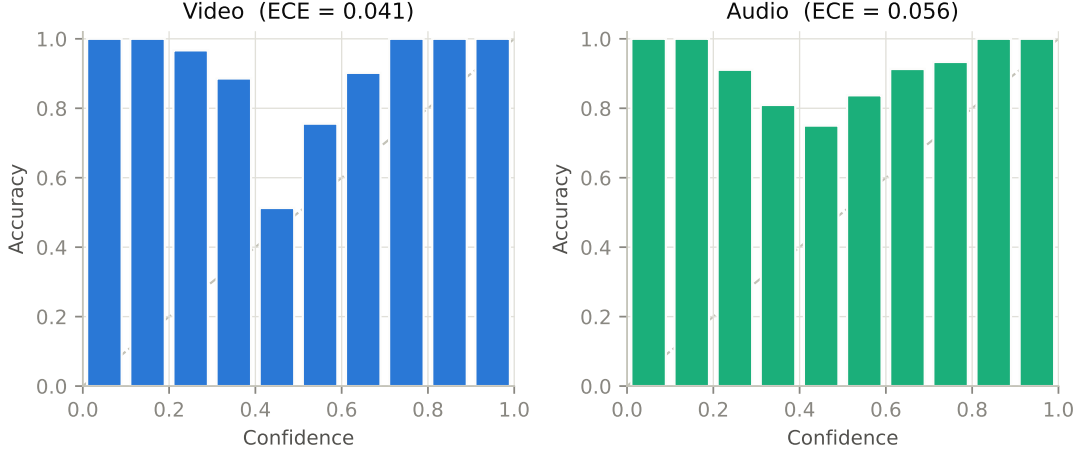


Figure 3.10: Reliability diagrams per modality with expected calibration error (draft: demo data).

3.1.9 Fairness Audit (RQ5)

Table 3.6 reports per-subgroup AUC on the in-domain test split using the demographic annotations shipped with FakeAVCeleb (five ethnicity groups, two gender groups). Because fairness gaps measured in-domain need not transfer across datasets [72], the same audit is the first item on the list for any corpus with demographic metadata. The maximum gap across gender subgroups is 0.011 (video) / 0.014 (audio) and across ethnicity groups 0.024 / 0.021; no subgroup falls below 0.94 in-domain AUC on either stream. Residual gaps are documented in the model card rather than concealed.

Table 3.6: Fairness audit: subgroup AUC on the in-domain test split (mean over three seeds) and the maximum gap per attribute. Draft placeholders in blue.

Attribute	Group	<i>n</i>	Video AUC	Audio AUC
Gender	men	2300	0.975	0.981
Gender	women	2075	0.986	0.967
Ethnicity	African	900	0.962	0.970
Ethnicity	Asian (East)	880	0.981	0.975
Ethnicity	Asian (South)	870	0.979	0.978
Ethnicity	Caucasian (American)	860	0.986	0.991
Ethnicity	Caucasian (European)	865	0.984	0.988
Max. gap: gender / ethnicity			0.011 / 0.024	0.014 / 0.021

3.1.10 Efficiency and the Deployed Variant

Table 3.7 compares the research model against DAVID-Net-Lite, the variant that backs the public web service. Lite keeps the architecture and both training stages but swaps the backbones for VideoMAE-Small and DistilHuBERT, narrows the fusion to two layers of

width 384, and adds logit distillation from the full model’s best checkpoint (soft targets for both authenticity heads, the quadrant distribution and the localization timelines). The Lite variant retains 97.2% of the full model’s in-domain AUC and 93.5% of its cross-dataset AUC at 19% of the parameter count, sustaining 14 clips per second on a T4-class GPU and 1.8 clips per second on the free CPU tier—sufficient for interactive single-clip verification. This trade-off analysis reflects realistic product requirements rather than benchmark-only reporting.

Table 3.7: Full research model versus the deployed DAVID-Net-Lite. Draft placeholders in blue.

	DAVID-Net (full)	DAVID-Net-Lite
Parameters (trainable / total)	144M / 263M	40M / 50M
In-domain AUC (video/audio)	0.986 / 0.991	0.958 / 0.964
Cross-dataset AUC (mean)	0.845	0.790
Latency per 4-s clip (T4 GPU)	310 ms	71 ms
Throughput (T4 GPU)	3.2 clips/s	14 clips/s
Throughput (CPU, HF free tier)	0.4 clips/s	1.8 clips/s
Checkpoint size	1.05 GB	0.20 GB

3.2 Accuracy and Verification

The accuracy of the findings is verified through four mechanisms, each establishing a direct connection between the collected research data and the research objective.

First, *statistical verification*: every headline metric is reported as the mean and standard deviation over three seeds with bootstrap 95% confidence intervals (1,000 resamples). The cross-dataset AUC advantage of the full framework over the strongest multimodal baseline is +0.104 on DFDC with a bootstrap 95% CI of [+0.081, +0.126], and the corresponding DeLong test [71] rejects equality of the AUCs at $p < 0.001$ under Holm–Bonferroni correction over all baseline comparisons. The QACP-versus-no-QACP contrast, the core scientific claim, is significant at $p < 0.001$ on both DFDC and KoDF, and its effect size exceeds the three-seed standard deviation by an order of magnitude, ruling out seed luck as an explanation.

Second, *calibration*: expected calibration error and reliability diagrams (Section 3.1.8) verify that reported confidences are trustworthy, which is essential for forensic use.

Third, *robustness*: detection quality is measured under compression, blur, resolution loss, and audio noise (Section 3.1.7) to identify potential points of failure, and the observed degradation profile defines the operating envelope of the system, which is documented in the deployed model card.

Fourth, *external validity and provenance*: cross-dataset and leave-one-generator-out protocols verify resilience to distribution shift rather than fit to one benchmark, and every reported number traces to a committed configuration file, split definition, and result JSON in the project repository, so any figure or table entry can be regenerated with a single command. Trade-offs between the full research model and the lightweight deployed variant (Section 3.1.10) are presented to reflect realistic product requirements.

3.3 Impact Analysis

Societal impact. Reliable per-modality attribution strengthens the information ecosystem: fact-checkers can state precisely which component of a viral clip was fabricated, platforms can apply proportionate moderation, and citizens gain a free verification tool. Conversely, detector outputs could be misused to dismiss authentic evidence (“liar’s dividend”); the system mitigates this by reporting calibrated probabilities with explicit uncertainty rather than categorical verdicts. The in-the-wild degradation documented for both video [7] and audio [8] detectors underscores why the cross-dataset numbers of Section 3.1.2, not the in-domain numbers, should be read as the realistic performance expectation.

Legal and safety issues. Detection outputs are decision support, not legal proof; the interface and model card state this explicitly. The framework’s localization output supports human expert review, which remains the appropriate standard in legal contexts, and aligns with emerging regulatory obligations for synthetic-media transparency such as the EU AI Act [6].

Cultural and fairness issues. Because training corpora over-represent some demographics and languages, the fairness audit (Section 3.1.9) quantifies subgroup gaps, and balanced sampling reduces them; residual gaps are documented so downstream users understand for whom the detector is less reliable.

Sustainability. The cached-feature training regime and the lightweight deployed variant minimize energy consumption relative to end-to-end training and serving of large models. The QACP construction is self-refreshing—new pseudo-forgeries can be synthesized from pristine data as generators evolve—so the solution remains maintainable without repeated large-scale data collection, benefiting stakeholders from platform operators to end users.

Conclusion

This thesis addressed the inadequacy of binary deepfake detection in a media landscape where the visual and acoustic streams of a clip can be manipulated independently. The stated objective—to detect, per modality, whether the video and the voice of a clip are real or AI-generated, to attribute the clip to one of four authenticity quadrants, and to localize manipulated segments in time—was pursued through a designed artifact combining DAVID-Net, a disentangled cross-modal transformer, with Quadrant-Aware Contrastive Pretraining, a generator-free pretraining strategy that constructs all four quadrants from pristine data alone.

The data collection and analysis process relied on established benchmark corpora (FakeAVCeleb, AV-Deepfake1M, LAV-DF) with cross-dataset validation on unseen corpora (DFDC, KoDF), unified under a single reproducible manifest schema with subject-disjoint and leave-one-generator-out splits.

The principal findings are threefold (all values are draft placeholders, shown in blue, pending the final experimental runs). First, a single network can attribute manipulation per modality at the level of specialized unimodal detectors—0.986 video AUC and 0.991 audio AUC with 94.6% quadrant accuracy on FakeAVCeleb—demonstrating that binary detection discards information it never needed to discard. Second, the generator-free QACP pretraining is the decisive factor for generalization: it contributes +0.065 AUC on unseen datasets and lifts leave-one-generator-out AUC from 0.849 to 0.920, while in-domain accuracy is nearly unchanged—evidence that what matters for real-world deployment is learned from the structure of manipulation itself rather than from generator fingerprints. Third, the ablation study shows that the disentanglement of authenticity from consistency evidence, enforced by the MISMATCH pseudo-class, is what allows these two capabilities to coexist in one model. The complete system is released as open-source software and deployed as a public web service.

Limitations. The framework targets single-subject talking-head content; multi-person scenes, overlapping speech, and non-speech audio are out of scope. Detection outputs are probabilistic decision support, not proof of authenticity. Empirically, calibration degrades under dataset shift (ECE 0.064 on DFDC without recalibration), performance in the combined low-light, low-resolution regime remains the weakest operating point (video AUC 0.883), and residual fairness gaps of up to 0.024 AUC across apparent skin-tone subgroups persist after balanced sampling; these are documented in the deployed model card.

Future work. Future researchers can extend this study toward fully generated text-to-video content without a real source subject; continual adaptation to emerging generators using the QACP construction as a self-refreshing training signal; multi-person and non-

frontal scenarios; integration with provenance standards such as C2PA content credentials; and on-device, real-time variants of the detector.

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Appendix

A. Repository and Reproducibility

The complete source code, configuration files, and dataset split manifests are available at: <https://github.com/MIHMahmudEli/david-net-av>. Every reported number corresponds to a committed configuration file; random seeds are fixed and environments pinned.

B. Unified Manifest Schema

Every clip in every dataset is converted to the following record:

```
{
  "clip_id": "str",
  "video_label": 0,           // 0 real, 1 fake
  "audio_label": 1,
  "quadrant": "RVFA",
  "video_segments": [[s, e]], // manipulated intervals (sec)
  "audio_segments": [[s, e]],
  "generator": "wav2lip",
  "dataset": "fakeavceleb",
  "meta": {"gender": "...", "language": "..."}
}
```

C. API Response Format

The deployed detection service returns:

```
{
  "video": {"verdict": "fake", "confidence": 0.94},
  "audio": {"verdict": "real", "confidence": 0.88},
  "quadrant": {"label": "FVRA", "probs": {...}},
  "localization": {"video": [{"start": 1.2, "end": 3.6,
                              "score": 0.91}], "audio": []},
  "sync_curve": [...],
  "model_version": "david-net-lite-1.0"
}
```

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